

What is it?

Last revised in March 2025

- Diabetes mellitus is a group of metabolic disorders in which persistent hyperglycaemia (random plasma glucose more than 11.1 mmol/L) is caused by deficient insulin secretion, resistance to the action of insulin, or both [[Mayer-Davis, 2018 \(/topics/diabetes-type-2/references/\)](#)] [[WHO, 2019 \(/topics/diabetes-type-2/references/\)](#)].
 - Insulin deficiency and insulin resistance lead to the abnormalities of carbohydrate, fat, and protein metabolism that are characteristic of diabetes mellitus [[Mayer-Davis, 2018 \(/topics/diabetes-type-2/references/\)](#)].
- Diabetes mellitus is classified as [[Mayer-Davis, 2018 \(/topics/diabetes-type-2/references/\)](#)] [[Zeitler, 2018 \(/topics/diabetes-type-2/references/\)](#)] [[WHO, 2019 \(/topics/diabetes-type-2/references/\)](#)] [[NICE, 2020a \(/topics/diabetes-type-2/references/\)](#)]:
 - Type 1 diabetes — an absolute insulin deficiency causes persistent hyperglycaemia (insulin activity is normal). See the CKS topic on [Diabetes – type 1 \(/topics/diabetes-type-1/\)](#) for more information.
 - Type 2 diabetes — insulin resistance and a relative insulin deficiency result in persistent hyperglycaemia.
 - Gestational diabetes — hyperglycaemia develops during pregnancy and usually resolves after delivery, although the woman is at increased risk for overt type 2 diabetes in the future.
 - Other specific types of diabetes, including:
 - Monogenic diabetes (caused by specific gene mutations, some affecting pancreatic beta-cell function, others affecting insulin action). The previously known 'maturity-onset diabetes of the young' presents with an autosomal dominant family history of diabetes, with typically mild, non-ketotic disease presenting in adolescence or early adulthood.
 - Secondary to conditions or diseases of the exocrine pancreas, such as pancreatitis, malignancy, trauma or surgery, cystic fibrosis, or haemochromatosis, for example.
 - Endocrine disorders, such as acromegaly, Cushing's syndrome, hyperthyroidism, or pheochromocytoma.
 - Drug- or chemical-induced, such as following corticosteroid treatment.
- The term 'type 2 diabetes' has replaced the terms 'non-insulin-dependent diabetes mellitus (NIDDM)' and 'adult-onset diabetes', as people with type 2 diabetes may require insulin, and type 2 diabetes is increasingly being diagnosed in children and young people.

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